Genus and Gonality of dynatomic/preimage curves

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1. Dynatomic/preimage curves.

Algebraic family: f_t, $t \in \Lambda$

Λ: (irr.) algebraic curve (var.)

endo. of $\mathbb{P}^N_{\mathbb{C}}$ (i.e. $N=1$)

$$f: \Lambda \times \mathbb{P}^N \dashrightarrow \Lambda \times \mathbb{P}^N \quad / \Lambda$$

$$(t, z) \longmapsto (t, f_t(z))$$

algebraic of degree $d \geq 2$.

Def: f is isotrivial if $f_t : \mathbb{P}^N \to \mathbb{P}^N$ are conjugate $\forall\, t \in \Lambda$. (i.e. $N=1$)

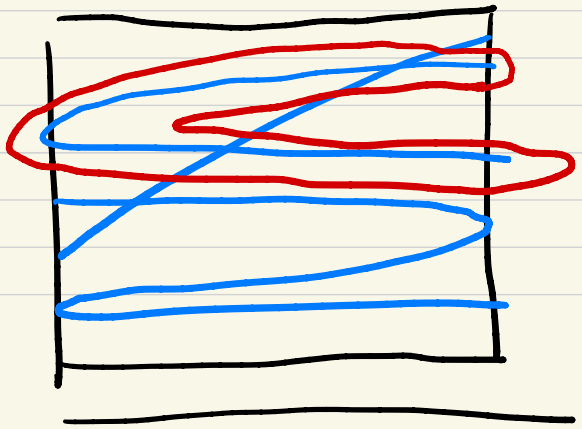
Dynatomic curves := "a family of preperiodic points of f".

$\forall\, n > m \geq 0$,

$$\{(t,z) \in \Lambda \times \mathbb{P}^N \mid f^n(t,z) = f^m(t,z)\}$$

defines a 1-dim closed subvar. in $\Lambda \times \mathbb{P}^N$.

i.e.



$\forall$ irr. component is a Dynatomic curve.

Marked point := a section of $\Lambda \times \mathbb{P}^N \to \Lambda$ i.e.

$$a : \Lambda \to \Lambda \times \mathbb{P}^N$$
$$t \mapsto (t, a_t)$$

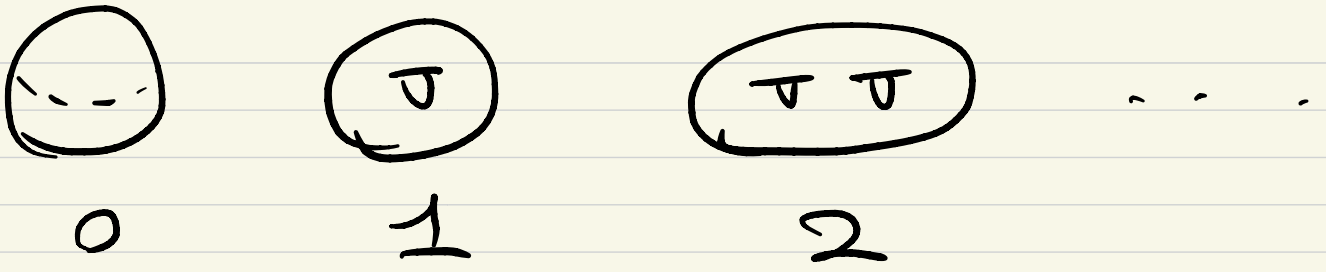
$$\Gamma_a := a(\Lambda) \subset \Lambda \times \mathbb{P}^N$$

Preimage curves := "family of points in backward orbits of a marked point"

$$\forall\, n \geq 0, \quad \{(t,x) \in \Lambda \times \mathbb{P}^N \mid f^n(t,x) = a_t\}$$ ← dim 1

$\forall$ irr. component is a Preimage curves (for a).

genus :



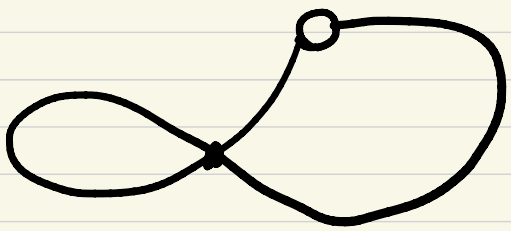
0 1 2 . . .

genality : minimal deg : $C \twoheadrightarrow \mathbb{P}^1$

e.g. hyperelliptic curves genality = 2

genus unbounded.

general algebraic curve



C

compactification + desingularization

$\tilde{C}$

Our results:

Thm 1 (J. - X.) Assume the family

$$f : \Lambda \times \mathbb{P}^1 \to \Lambda \times \mathbb{P}^1$$

is non-isotrivial.

Let Γ_n, $n \geq 1$, distinct dynatomic/preimage curves

Then • $\operatorname{gon}(\Gamma_n) \to +\infty$

• Not flexible Lattès $\iff$ $\dfrac{\operatorname{genus}(\Gamma_n)}{\deg(\Gamma_n \to \lambda)} \to +\infty$

$\Rightarrow$ Gonality / genus Conj.

Ex: Pick $F: \mathbb{P}^1 \to \mathbb{P}^1 / \mathbb{C}$ $\deg F = d \geq 2$.

$f = id \times F : \Lambda \times \mathbb{P}^1 \to \Lambda \times \mathbb{P}^1$

the trivial family i.e. $f_t = F \quad \forall t \in \Lambda$

$\forall x \in Per(F)$, $\Lambda \times \{x\} \times \{x\}$ is a dynatomic curve. (similar for preimage curves.)

$\Rightarrow$ The non-isotriviality assumption in Thm 1 is necessary.

Rem: (Ji-X.) Have N-dim version under mild condition.

DUBC (1-dim)

Dynamical "Uniform Boundedness Conj" (Levin, Morton-Silverman)

$\forall \; d \geq 2, \; D \geq 1, \; \exists \; B = B(d, D, k)$

$\forall \; f : \mathbb{P}^1 \to \mathbb{P}^1 \; / k, \; \deg f = d,$

- $\#\left\{ y \in O_f^{-n}(0) \;\middle|\; [k(y) : k] \leq D \right\} \leq B$ (Pre-image)

(0 can be any point)

- $\#\left\{ y \in \mathrm{Prep}(f) \;\middle|\; [k(y) : k] \leq D \right\} \leq B$ (Preperiodic points)

UBC : $\exists\ B = B(g, D, K)$ s.t.

$\forall\ A$ abelian var. $/K$

$$\#\{y \in A_{tor} \mid [K(y) : K] \leq D\} \leq B.$$

(Morton-Silverman, Fakhruddin) DUBC (∀ dim) $\Longrightarrow$ UBC

Observation : $[2]: A \to A$

$x \mapsto 2x$

Then $A_{tor} = P_{prep}([2])$.

Reformulation

DUBC (1-dim) $f: \mathbb{A}^1 \times \mathbb{P}^1 \to \mathbb{A}^1 \times \mathbb{P}^1 \ / \mathbb{A}^1 \quad \deg = d \geq 2$

algebraic family over $K = \#$ field.

$s: \mathbb{A}^1 \to \mathbb{P}^1$ a section $/K$

Then $\forall\, D \geq 1$, $\exists\, B = B(D)$ s.t.

- $\#\left\{ (t,y) \in (\mathbb{A}^1 \times \mathbb{P}^1)(K) \;\middle|\; y \in O_{f_t^{-1}}(s(t)) \right\} \leq B$
- $\#\left\{ (t,y) \in (\mathbb{A}^1 \times \mathbb{P}^1)(K) \;\middle|\; y \in \mathrm{Prep}(f_t) \right\} \leq B$

Reformulation

"$f: M_d \times \mathbb{P}^1 \circlearrowright$ tautological family"

DUBC (1-dim) $f: \Lambda \times \mathbb{P}^1 \to \Lambda \times \mathbb{P}^1 / \Lambda$ $\deg = d \geq 2$

algebraic family over K = # field.

$s: \Lambda \to \mathbb{P}^1$ a section /K

Then $\forall D \geq 1$, $\exists B = B(D)$ s.t.

- $\#\{(t,y) \in (\Lambda \times \mathbb{P}^1)(K) \mid y \in O_{f_t^{-1}}(s(t))\} \leq B$
- $\#\{(t,y) \in (\Lambda \times \mathbb{P}^1)(K) \mid y \in \mathrm{Prep}(f_t)\} \leq B$

Thm 2 (Ji-X.)

Assume $\dim \Lambda = 1$. Then (1-dim)-DUBC holds for Pre-images.

Rem: 1. (Ji-X.) When $\dim \Lambda = 1$, (N-dim)-DUBC holds under mild condition. for Pre-images.

↑ Bifurcation theory.

2. (Ji-X.) When $\dim \Lambda = 1$, above holds for preperiodic points when K = function field.

Rem 1). Thm 1 $\Rightarrow$ Thm 2 (Frey Lem based on Mordell-Lang solved by Faltings)

2). Thm 2 have N-dim version under mild conditions.

3). { dynatomic curves, preimage curves } ← "small height"

We have general version for small height curves.

Previous results:

moduli space dim 1

UBC for abelian var.: dim 1 (elliptic curve) ↑

Mazur 77, 78: $/\mathbb{Q}$ Merel 96: # field.

dim ≥ 1: Cadoret–Tamagawa 12: ℓ-primary torsions in $A(K)$ over 1-dim family. (Gabis reps.)

Ji–Song–X.: New geometric prove.

Function field: $K = \mathbb{C}(B)$ B curve.

Yap Jit Wu – Looper (26) ← (Szpiro Conj = abc Conj.)

Gao – Gu (in progression) Quantitative version, new proof

DUBC for preimages

#field. All 1-dim, 1-parameter

(1) Faber-Hutz-Ingram-Jones-Manes-Tucker-Zieve 09:

$f_t(z) = z^2 + t$ preimages of $a \in \mathbb{P}^1(K)$.

(2) Ingram12: Lattès family ← elliptic curves.

(3) $f_t(z) = z^d + t$, $d \geq 2$

explicit bounds.

Faber 10, Faber-Hutz-Stoll 11, Hutz-Hyde-Krause 11, Sano 25

DUBC

Looper 21: DUBC holds for 1-dim poly. assuming abc/abcd Conj.

Conality conj. $\approx$ GUBC one parameter

(1) Dujle-Poonen 20. $f_t(z) = z^d + t$

(2) Nguyen-Saito 96. flexible Lattès. $\leftarrow$ elliptic curve

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Our proof: (Thm 1)

① Woven currents

② Arithmetic equidistribution
(Yuan, Moriwaki, Yuan-Zhang)

③ Bifurcation theory.

Bifurcation theory : | 1 dim : Mané - Sad - Sullivan, Lyubich, DeMarco

| ≥ 2 dim : Berteloot - Bianchi - Dupont

$$\forall\, t \in \Lambda, \quad \mathrm{Lyap}(f_t) := \int_{\mathbb{P}^k} \log \det |df| \, \mu_{f_t}$$

Not depend on the metric — maximal entropy measure.

The function $\mathrm{Lyap} : \Lambda \longrightarrow \mathbb{R}$

$$t \longmapsto \mathrm{Lyap}(f_t)$$

is a p.s.h function (Hölder continuous)

$\Rightarrow T_{bif} := dd^c \mathrm{Lyap}$ is a positively closed $(1,1)$-current on Λ.

Def: $Bif(f) := \operatorname{Supp} T_{bif} \subset \Lambda$

↑ Bifurcation locus of f

$$Stab(f) := \Lambda \setminus Bif(f)$$

Stable locus of f.

So $\Lambda = Bif(f) \sqcup Stab(f)$

↑ closed

↑ open

$f_t, t \in \Lambda$ is called **stable** if $\mathrm{Bif}(f) = \emptyset$.

Stablity $\Longleftrightarrow$ Bigness of a natural adelic line bundle

Yuan-Zhang

Ex: 1. f isotrivial $\Rightarrow$ f stable.

2. $n=1$, f **Lattès family**:

family of elliptic curve $\rightsquigarrow$

$$\begin{array}{ccc} E_t & \xrightarrow{\times l} & E_t \\ \downarrow & & \downarrow \\ E_t/\pm 1 \simeq \mathbb{P}^1 & \xrightarrow{f_t} & \mathbb{P}^1 \end{array}$$

$|l| \geq 2$

Thm (McMullen): $N=1$, $f_t, t\in\Lambda$ is stable iff either f is isotrivial or f is a Lattès family.

Question: Classify stable families when $N\geq 2$?

$\Longleftrightarrow$ "Positivity of adelic line bundle".

Bifurcation theory + Yuan-Zhang $\Longrightarrow \exists\, U\subset \mathcal{M}_d^N$ Zar. open dense s.t. $f_t, t\in\Lambda$ not stable if $\Lambda\cap U\neq\emptyset$.

$\Longrightarrow$ Very few stable families.

Proof of Thm 1 (genus version)

For simplicity assume $f: \Lambda \times \mathbb{P}^1 \to \Lambda \times \mathbb{P}^1 / Q$

By contradiction: assume

$$\frac{\text{genus}(\Gamma_n)}{\deg(\Gamma_n \to \Lambda)} \leq C \quad (*)$$

Step 1: Define $O(\Gamma_n) = Gal(\overline{\mathbb{Q}(B)}/\mathbb{Q}(B))$

orbit of Γ_n.

Taking subsequence, may assume

$$\frac{[O(\Gamma_n)]}{\deg_{\pi_1}(O(\Gamma_n))} \longrightarrow S$$

in the sense of current.

Reason: "Preperiodic/preimage points are small".

$$\forall\, t \in \Lambda\,, \quad R_t := [\pi_1^{-1}(t)]$$

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$\mathbb{P}^1$

Relative arithmetic equidistribution Thm (Morimaki)

$\Longrightarrow$ For $t \in \Lambda$, if

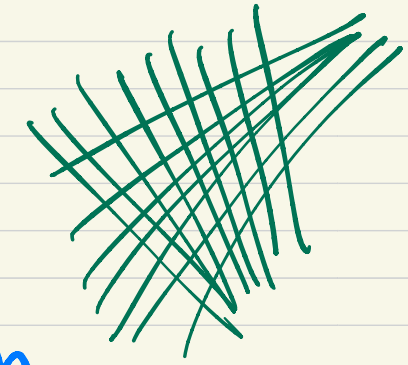
$$S \wedge R_t$$ "well-defined",

then $S \wedge R_t = \mu_{f_t}$ ← maximal entropy measure on $\pi_1^{-1}(t)$.

Step 2 (Woven currents)

A bidim $(1,1)$ current T is called uniformly Woven if

$$T = \int [\Gamma_a] \, d\mu(a)$$



$[\Gamma_a]$: closed (analytic) curves

μ: positive measure on "space of curves"

By $\circledast \Rightarrow S$ in step 1 is a strongly approximable Woven current

= "Well approximated by uniformly Woven current"

Property: $\exists\ S_m, m \geq 0$ uniformly Woven current on open subsets Ω_m

s.t. (1) $S_m \leq S$;

(2) $(S - S_m) \wedge T \to 0$

$\forall$ positively closed $(1,1)$-current having continuous potential

Step 3 (Conclude by bifurcation theory).

McMullen's rigidity
+
Lattès case (Nguyen-Saito)
} ⇝ Assume f not Stable.

key: ergodic theory

Step 1 + Step 2 $\Rightarrow$ We can find a parameter $t_0 \in \mathrm{Bif}(f)$, a disk $\mathbb{D} \ni t_0$ and $\forall$ repelling periodic point x of f_{t_0}

"move holomorphically".

Holomorphic move: $\exists \quad p: \mathbb{D} \to \mathbb{P}^1$

s.t. (1) $p(t_0) = x$

(2) $\forall t \in \mathbb{D}$, $p(t)$ is a repelling periodic pt of f_t.

$\Longrightarrow$ f is stable at t_0.

1-dim Bifurcation theory.

□

Contradiction!

($t_0 \in Bif(f)$)

Thank yan!